Patient Health Record Consent Management System

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Lab 3: Component Modelling & Architectural Pattern Selection

Architecture Selection: Layered Architecture

Justification

We selected Layered Architecture for the Patient Health Record Consent Management System to organise its user interface, consent-processing logic, verification mechanisms, and record-storage operations into separate components.

Reason 1 — Time-Bounded Permission Management

Patients must be able to grant access to specific diagnostic records for a defined period. The Consent Management Service stores the permitted record scope, recipient, status, and expiry time. Keeping this functionality separate allows consent rules to be managed consistently throughout the system.

Reason 2 — Easier Integration and Maintenance

The Health Record Gateway communicates with the Health Record Store through a defined interface, while the Identity & Access Verification Service handles identity checks. These boundaries simplify maintenance and make it easier to integrate additional clinics or diagnostic laboratories without redesigning every component.

Security Advantage

The system validates the healthcare professional’s identity and checks the patient’s consent before returning medical records. An append-only audit trail records consent changes and access decisions, supporting accountability and helping identify unauthorized access attempts.

Performance Benefit

Separating authorization from record retrieval allows the system to reject invalid requests before querying the Health Record Store. This reduces unnecessary operations and supports the specified target of completing at least 95% of authorization requests within two seconds under the required simulated load.